

MUHAMMAD SHAHRUKH KHAN

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PROFESSIONAL SUMMARY

Final-year Computer Science student who has shipped production-grade AI, full-stack, and offline-first systems across four technical internships, achieving up to 91% model accuracy and 30% faster inference. Currently building SkillLoom, an AI-driven hiring platform for Pakistan's 15M-worker textile sector, and Z-Stock, a multi-tenant offline-first inventory management system for Pakistan's retail sector. Comfortable across the full SDLC in Agile team environments.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL, Assembly, Rust
Frameworks/Libraries: FastAPI, Flutter, React.js, NumPy, Pandas, Scikit-learn, TensorFlow
AI / ML: Machine Learning, LLM Integration, Prompt Engineering, InsightFace, Computer Vision, KDE, Cosine Similarity, Claude API Integration, Generative AI
Backend & Data: RESTful APIs, PostgreSQL, Supabase, MongoDB, Database Design
DevOps & Tools: Docker, Nginx, systemd, Linux VPS Administration, Git, GitHub, CI/CD, Postman, Agile/Scrum, OOP, Data Structures & Algorithms

EDUCATION

Bachelor of Science in Computer Science *Expected: May 2027*
COMSATS University Islamabad, Wah Campus — AI, Software Engineering, Data Science, Cybersecurity

PROFESSIONAL EXPERIENCE

AI/ML & DevOps Intern | Pakistan Ordnance Factories (POF) *July 2026 – August 2026*

- Built and deployed a full-stack React/TypeScript frontend and FastAPI backend, then migrated the backend from managed platforms (Vercel/Railway) to a self-managed Linux VPS using Nginx, systemd, and Unicorn/Uvicorn workers for persistent, auto-restarting service.
- Hardened server security (SSH key auth, non-root sudo user, ufw firewall) and configured a custom domain with DNS records and Let's Encrypt SSL, taking the project from local development to a production-grade, publicly accessible deployment.
- Studied applied AI/ML engineering concepts (ML lifecycle, model evaluation, MLOps/CI-CD, model-serving APIs), framing the deployment and security work above as the production foundation an AI system needs to ship reliably.

AI & ML Engineering Intern | DevelopersHub Corporation *June 2025 – July 2025*

- Awarded Best Performance recognition, ranking in the top 5% of the intern cohort for contributions across AI/ML engineering projects.
- Built and deployed ML models with Python, Scikit-learn, NumPy, and Pandas, achieving up to 91% classification accuracy on structured datasets.
- Built and optimized FastAPI endpoints to serve model predictions, cutting average inference response time by ~30%.
- Applied feature engineering and model evaluation (cross-validation, ROC-AUC) within Agile sprint workflows, collaborating via Git and daily code reviews.

Python Programming Intern | Cosmicode Technologies *July 2025 – October 2025*

- Contributed to backend development using Python, FastAPI, and PostgreSQL in a 4-month Agile/Scrum engagement.
- Designed RESTful API modules with authentication, input validation, and error handling, improving system reliability.
- Optimized SQL queries and schemas, reducing query execution time by ~25% on high-frequency endpoints.

Python Programming Intern | CodeAlpha *June 2025*

- Received a Letter of Recommendation for analytical skills, rapid technology adoption, and consistent technical contributions.
- Built Python automation and data-processing scripts with NumPy and Pandas, cutting manual processing time by an estimated 40%.

KEY PROJECTS

SkillLoom — AI-Powered Textile Workforce Hiring Platform
Final Year Project · Python, FastAPI, Flutter, React.js, Supabase, PostgreSQL, Docker, InsightFace

- Architected a full-stack hiring platform for Pakistan's informal textile sector, spanning 15M+ potential users, from requirements through deployment.
- Engineered AI job-matching using cosine similarity for skills ranking and the Haversine formula for location-based recommendations.
- Implemented InsightFace-based biometric verification, removing text-based login barriers for low-literacy workers.
- Built KDE-based demand heatmaps for real-time workforce planning.
- Delivered a Flutter worker app and React.js employer portal with JWT authentication and Docker-based deployment.

Z-Stock (Inven) — Multi-Tenant Offline-First Inventory Management System
Python, FastAPI, PostgreSQL, SQLAlchemy, Tauri, Rust, React, TypeScript, Docker

- Architected a multi-tenant, offline-first inventory management system for Pakistan's retail/wholesale market as a monorepo: FastAPI/PostgreSQL backend, Tauri (Rust + React) offline POS client, and React/TypeScript web-admin portal.
- Implemented offline-first POS sync using local SQLite (WAL mode) with a sync outbox, allowing till operations to continue through network outages and reconcile with the central ledger on reconnect.
- Built core inventory logic covering batch/FEFO tracking, audit logging, and discount governance, backed by 60+ automated test suites.

PMNH — QR-Based Mobile Museum Guide
Commissioned Client Project · Python, FastAPI, Flutter, Supabase, PostgreSQL

- Delivered an end-to-end backend for the Pakistan Museum of Natural History, digitizing visitor experience via a WiFi-gated, QR-driven mobile architecture.
- Built an admin portal enabling non-technical museum staff to manage exhibits, multimedia content, and visitor analytics without developer involvement.
- Coordinated network-level access control and APK sideload distribution with client IT, delivering Phase 1 MVP on schedule and ready for on-site visitor use.

Additional Projects: CRM — owner-scoped CRM with contacts, pipelines, deals, tasks, quotes, and reporting (React, Express, tRPC, Drizzle, MySQL/TiDB); AI-Review-Pro — AI code reviewer scoring security, readability, and complexity (Next.js, Supabase, Claude API).

CERTIFICATIONS

- Generative AI Boot Camp — Coursera, 2024
- Python for Everybody — Coursera / University of Michigan, 2024
- Database Fundamentals (SQL & MongoDB) — Coursera / IBM, 2024

Languages: English (Professional Fluency) • Urdu (Native)